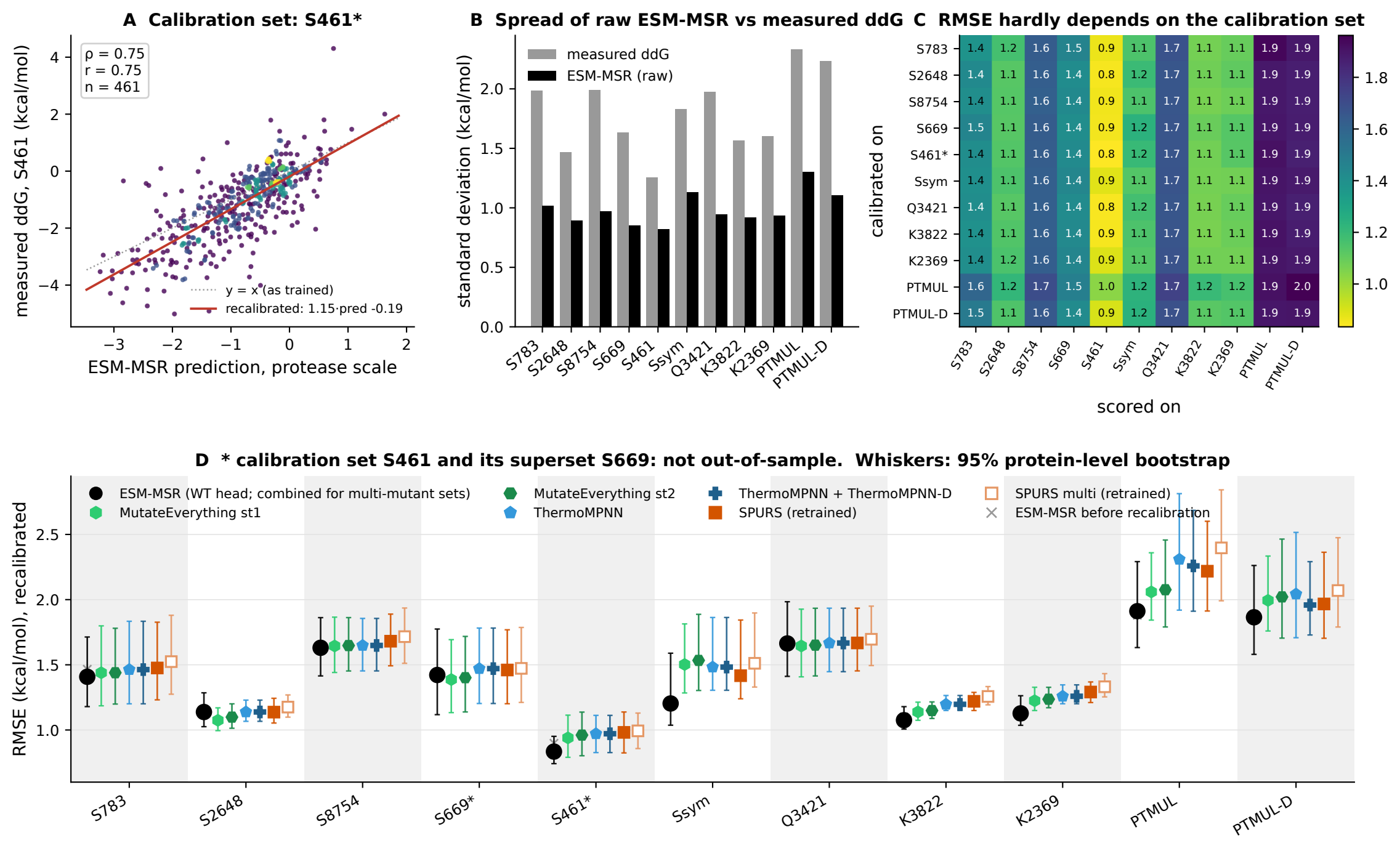

A Calibration set: S461*
ρ = 0.75
r = 0.75
n = 461
y = x (as trained)
recalibrated: 1.15·pred -0.19
measured ddG, S461 (kcal/mol)
ESM-MSR prediction, protease scale
B Spread of raw ESM-MSR vs measured ddG
measured ddG
ESM-MSR (raw)
standard deviation (kcal/mol)
S783 S2648 S8754 S669 S461 Ssym Q3421 K3822 K2369 PTMUL PTMUL-D
C RMSE hardly depends on the calibration set
calibrated on
S783 1.4 1.2 1.6 1.5 0.9 1.1 1.7 1.1 1.1 1.9 1.9
S2648 1.4 1.1 1.6 1.4 0.8 1.2 1.7 1.1 1.1 1.9 1.9
S8754 1.4 1.1 1.6 1.4 0.9 1.1 1.7 1.1 1.1 1.9 1.9
S669 1.5 1.1 1.6 1.4 0.9 1.2 1.7 1.1 1.1 1.9 1.9
S461* 1.4 1.1 1.6 1.4 0.8 1.2 1.7 1.1 1.1 1.9 1.9
Ssym 1.4 1.1 1.6 1.4 0.9 1.1 1.7 1.1 1.1 1.9 1.9
Q3421 1.4 1.1 1.6 1.4 0.8 1.2 1.7 1.1 1.1 1.9 1.9
K3822 1.4 1.1 1.6 1.4 0.9 1.1 1.7 1.1 1.1 1.9 1.9
K2369 1.4 1.2 1.6 1.4 0.9 1.1 1.7 1.1 1.1 1.9 1.9
PTMUL 1.6 1.2 1.7 1.5 1.0 1.2 1.7 1.2 1.2 1.9 2.0
PTMUL-D 1.5 1.1 1.6 1.4 0.9 1.2 1.7 1.1 1.1 1.9 1.9
scored on
D * calibration set S461 and its superset S669: not out-of-sample. Whiskers: 95% protein-level bootstrap
ESM-MSR (WT head; combined for multi-mutant sets)
MutateEverything st1
MutateEverything st2
ThermoMPNN
ThermoMPNN + ThermoMPNN-D
SPURS (retrained)
SPURS multi (retrained)
ESM-MSR before recalibration
RMSE (kcal/mol), recalibrated
S783 S2648 S8754 S669* S461* Ssym Q3421 K3822 K2369 PTMUL PTMUL-D